ReconciCA: Auditable GST Purchase Reconciliation with Explicit Ambiguity

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***Abstract*—Businesses compare their purchase registers with India's auto-drafted GSTR-2B statement before assessing Input Tax Credit. Invoice-number errors, date differences, and amount discrepancies make first-hit spreadsheet lookups unsafe: plausible records can be paired silently. We present ReconciCA, a browser-based system whose deterministic Dart engine groups records by supplier GSTIN, compares invoice identity, date, and five tax components, and assigns each source document at most once. Exact ties become explicit ambiguous decisions. An optional Purchase Compiler reads bill images through a vision-language model, applying a fail-closed validation gate and human review before producing register rows. Evaluation on an author-labelled corpus of 50 invoice-number pairs gives held-out precision 0.765, recall 1.000, and F1 0.867 at threshold 0.80. These figures measure invoice-number similarity within this corpus, not production accuracy. Across 26 generated scenarios, shuffling input order did not change decisions. We also identify a counterexample in which the same-date-first tier order can produce a false match. All reconciliation inputs and decisions remain in browser memory; the optional bill-reading path sends selected files to a third-party inference service.**

***Index Terms*—GST reconciliation, GSTR-2B, input tax credit, record linkage, ambiguity, auditability, invoice extraction**

I. Introduction

Under India's Goods and Services Tax, Input Tax Credit (ITC) depends on statutory conditions, including the supplier-furnished invoice details communicated to the recipient [1]. GSTR-2B is the portal's auto-drafted ITC statement; the Invoice Management System (IMS) lets recipients act on supplier records before filing [2], [3]. Before taking action, a business must determine whether each portal invoice corresponds to one in its own Purchase Register.

This is a record-linkage problem with consequential operational errors. Invoice numbers may contain transcription errors, dates may differ, and amounts may be rounded. A simplistic first-fit lookup can silently pair incorrect records, obscuring a missing invoice. ReconciCA emphasizes repeatable decisions, exact integer-money comparisons, data provenance, and explicit ambiguity. The matching engine is deterministic and operates entirely inside the browser. A separate Purchase Compiler assists users without an existing register by extracting bill data for human review.

A. Motivation

The system aims to deliver consistent decisions regardless of input order, eliminate floating-point comparison errors, explain every reconciliation outcome, prevent duplicate invoice assignments, and retain reconciliation files on the user's device. Furthermore, it explicitly exposes uncertainty rather than applying arbitrary tie-breaking logic when candidate evidence is equal.

B. Key Contributions

This work contributes: (1) a tax-component-aware matching cascade featuring GSTIN blocking and same-date-first candidate generation; (2) a deterministic one-to-one assignment that isolates exact competing candidates as ambiguous; (3) an auditable browser application providing source-row citeback; and (4) an evaluation detailing measured performance alongside a counterexample to the cascade's tier order. The optional compiler is presented as a system extension rather than a validated extraction algorithm.

II. Related Work

A. Record Linkage

Record linkage identifies entries referring to the same real-world entity across disparate datasets. Recent literature categorizes these techniques into rule-based, probabilistic, and machine-learned approaches, frequently employing candidate blocking to lower pairwise comparison costs [4]. ReconciCA utilizes an interpretable rule cascade blocked on exact supplier GSTINs; its goal is conservative operational review rather than learned entity resolution or globally optimal matching.

B. Comparing Strings

Levenshtein distance measures edit operations required to transform one string to another [5]. Jaro and Jaro–Winkler metrics were additionally evaluated on our test corpus [6].

This paper does not propose a novel string metric. Recent document-understanding benchmarks explore vision-language models on visual documents, providing context for our optional compiler without validating its extraction accuracy [10].

TABLE I
RECONCILIATION OUTCOMES

Outcome	Interpretation
GREEN	Matched under the configured rules
AMBER	Candidate requires review: discrepancy or ambiguity
RED	Purchase Register document not found in GSTR-2B
PURPLE	GSTR-2B document not found in Purchase Register

C. GST Reconciliation Tools

The contribution of this work lies beyond basic portal-to-register matching. We focus on deterministic tie resolution, input-order invariance, component-level monetary comparisons, and auditable browser-local workflows.

D. Reading Invoices with Large Models

The Purchase Compiler uses the GLM-4.6V vision-language model to read bills. Model responses are treated as untrusted data and strictly gated prior to export. General model benchmarks do not constitute evidence of domain-specific invoice extraction accuracy [7].

E. Research Gap

No directly comparable published evaluation of input-order invariance and explicit exact-tie resolution was identified for GST invoice reconciliation in the reviewed literature.

III. PROBLEM FORMULATION

Let G represent GSTR-2B invoices and P represent Purchase Register rows. Each record contains a supplier GSTIN, invoice number, date, taxable value, and component amounts (IGST, CGST, SGST, cess). Purchase rows sharing a normalized GSTIN and invoice number are folded into a single document object. Each document receives exactly one decision, enforcing standard one-to-one cardinality while preserving source-row identifiers. Scope is restricted to B2B invoices within a single tax period; amendments, credit/debit notes, ISD credit, and imports are excluded.

IV. THE RECONCILIATION ENGINE

Fig. 1 illustrates the overall user and data flow. Parsing, normalization, matching, and visualization run entirely within the Flutter web client. The core Dart engine has no network or filesystem permissions. The optional compiler is the sole component that transmits user-selected bill files to an external inference endpoint.

A. Cleaning, Folding, and Money

GSTINs and invoice numbers are trimmed and converted to uppercase; consecutive whitespace is collapsed while punctuation and leading zeros are preserved. Multi-rate lines sharing an invoice identity are summed component-wise with provenance tracking. If conflicting dates occur across purchase rows for an invoice, the earliest date is deterministically selected and a diagnostic flag is emitted. Monetary amounts are converted to integer paise to eliminate floating-point precision issues.

B. GSTIN Check Character

GSTIN checksum failures produce data-quality diagnostic flags rather than match rejections. This prevents suppressing genuine matches purely due to malformed supplier identifiers.

C. Blocking and the Two-Tier Date Search

Exact GSTIN blocking eliminates cross-supplier comparisons, dramatically scaling down candidate pair evaluations [4]. Within each supplier block, the engine evaluates same-date candidates first. It then considers residual unassigned documents across a non-zero offset window of ± 30 days. While deterministic, this priority scheme introduces a operational risk: a same-date fuzzy match can prematurely consume a document that would otherwise form an exact cross-date match.

D. Comparing Invoice Numbers

For invoice strings a and b, similarity is defined as:

$$s(a,b) = 1 - \frac{d(a,b)}{\max(|a|,|b|)} \tag{1}$$

where $d(a,b)$ is the Levenshtein distance computed via dynamic programming [5]. Identical strings bypass calculation. The configured fuzzy similarity threshold is set to 0.80. String similarity alone is insufficient for match approval; date and tax component conditions must also pass.

E. Comparing Amounts

Let g and p represent component amounts in paise for GSTR-2B and Purchase Register records, respectively. The exact match condition requires $|g - p| \leq 1$, while relative tolerance demands:

$$10{,}000 \cdot |g - p| \leq 50 \cdot |g| \quad (0.5\%) \tag{2}$$

All tax components must satisfy these bounds simultaneously.

F. The Decision Cascade

Candidates are categorized sequentially based on predefined criteria. Stage 1 demands exact invoice numbers and identical dates within 1-paisa component bounds. Stage 2 evaluates fuzzy numbers ($s \geq 0.80$) on identical dates within 0.5% amount bounds (flagged as low-confidence). Same-date pairs with amount conflicts, as well as cross-date pairs within 30 days, are routed to AMBER status. Unmatched GSTR-2B records yield RED decisions, while unmatched Purchase Register rows yield PURPLE decisions.

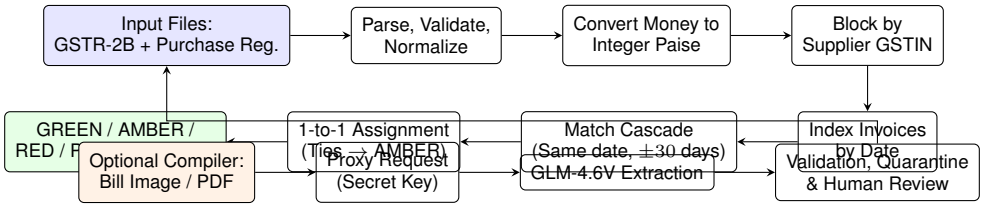


Fig. 1. ReconciCA system flow. The bill-reading path is optional and external; core reconciliation stays entirely within browser memory.

TABLE II
CANDIDATE RULES WITHIN THE SAME GSTIN

Stage	Candidate Condition	Status
1	Same date, exact number; Δamount $\leq$ 1p	GREEN
2	Same date, $s \geq 0.80$; Δamount $\leq 0.5\%$	GREEN*
3b	Same date, plausible number; amount conflict	AMBER
3	Date gap $\leq$ 30 days; plausible number	AMBER
4/5	No counterpart assigned	RED / PURPLE

Status key: GREEN = matched; GREEN* = low-confidence fuzzy match; AMBER = review/ambiguous; RED = GSTR-2B item unmatched; PURPLE = Purchase Register item unmatched.

G. One-to-One Assignment with Explicit Ambiguity

Candidate pairs are sorted descending by similarity, ascending by component monetary delta, and ascending by date offset. Assignments are accepted greedily provided neither document has been consumed. For identical-ranked candidates, connected components are built over the candidate graph. Graphs containing multiple competing edges are converted directly to explicit AMBER ambiguity outcomes.

H. A Failure of the Tier Order

The tier design exhibits a structural counterexample: a same-date fuzzy pair can be accepted prior to assessing a cross-date exact match. In testing, this produced a false GREEN match while leaving the genuine invoice unmatched. Thus, input-order invariance does not guarantee overall matching correctness.

V. THE PURCHASE COMPILER

The Purchase Compiler generates candidate Purchase Register entries from visual bill inputs. It detects MIME types, forwards documents through a key-secured proxy to GLM-4.6V, parses output schemas, and applies validation rules before human sign-off [7], [8].

A. Checking the File

Binary headers are validated against supported image and PDF magic numbers prior to transmission, rejecting mismatched extensions.

B. Reading the Bill

The client transmits schema constraints and file bytes. Credentials remain safely inside the server proxy, ensuring documents are transmitted only upon explicit user invocation.

C. The Admission Check

Exporting requires mandatory invoice IDs, dates, supplier GSTINs, and component sums. Discrepancies between declared totals and calculated component sums trigger quarantine flags requiring manual intervention.

D. Keeping Human Corrections Safe

Edit-provenance records distinguish model predictions from human overrides. Concurrent edits and re-uploads are handled using revision tags and content-based identifiers.

VI. SUPPLIER-SCOPED REPORTS

ReconciCA generates targeted vendor PDFs containing only relevant RED and AMBER items. PURPLE entries are retained internally as buyer-side register anomalies.

VII. EVALUATION

A. Data and Method

Evaluation relies on 45 synthetic scenarios alongside 50 author-labelled invoice-number pairs (33 typos, 17 distinct invoices), partitioned into 30 tuning and 20 held-out pairs. Confidence bounds use Wilson score intervals for proportions and percentile bootstrap methods for F1 scores [9].

TABLE III
HELD-OUT INVOICE-NUMBER RESULT ($N = 20$)

Measure	Estimate	95% Interval
Precision	0.765	[0.527, 0.904]
Recall	1.000	[0.772, 1.000]
F1	0.867	[0.714, 0.971]

B. Matching Results

At threshold 0.80, the held-out dataset yielded 13 TP, 4 FP, 0 FN, and 3 TN, achieving Precision 0.765, Recall 1.000, and F1 0.867. The presence of false positives demonstrates that string proximity alone cannot reliably prove invoice identity.

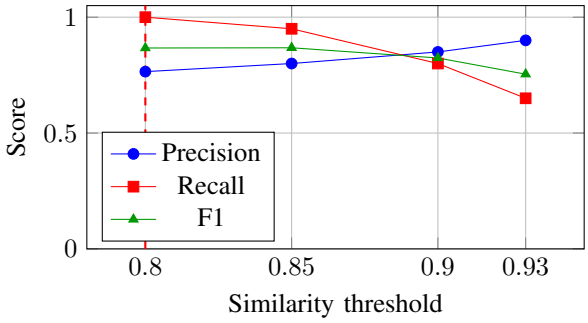


Fig. 2. Precision, recall, and F1 across thresholds (0.80–0.93). The dashed line highlights the optimal tuning threshold (0.80).

C. Analysis of Labelled Patterns

Evaluating edit patterns shows string ambiguity: 1-digit gaps at serial offsets of 1 or 10 were categorized as typos in 4 instances and different invoices in 1 instance. All 8 pairs with gap offsets ≥ 100 represented separate invoices.

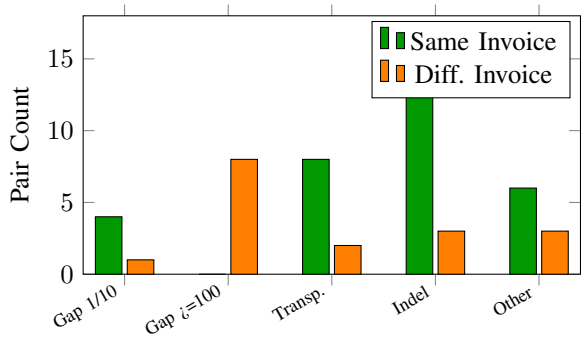


Fig. 3. Labelled pair counts by edit pattern.

D. Comparing Similarity Measures

Across the 50 pairs, Jaro distance achieved superior ROC-AUC relative to standard edit distance ratio (0.921 vs. 0.701), while Jaro–Winkler attained the highest held-out F1 score (0.897).

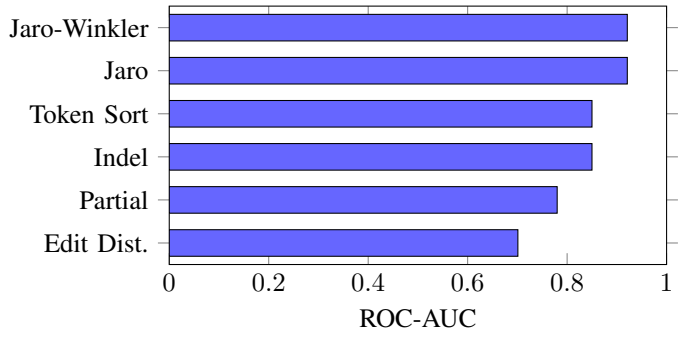


Fig. 4. ROC-AUC performance across string similarity metrics.

E. Independence from Input Order

Evaluating 26 scenarios across 25 random shuffles per scenario (650 iterations total) resulted in zero decision shifts. Disabling tie-grouping logic caused decision instability in 11 of 25 shuffles for tied inputs.

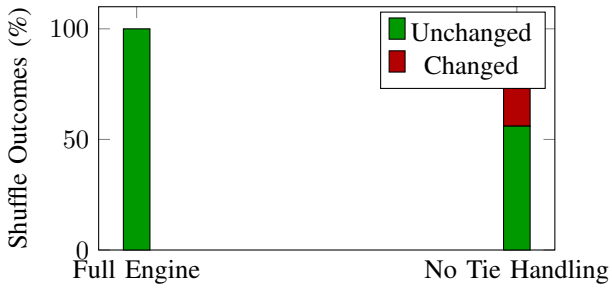


Fig. 5. Order-invariance stability test under input shuffling.

F. Speed and Test Coverage

For 2,200 documents per side across 28 dates, the Dart engine achieved a median execution time of 608 ms on Apple M4 Pro hardware. The test suite contains 609 automated unit and integration tests across engine and client modules.

VIII. Conclusion

ReconciCA pairs GSTIN blocking with tax-component rules and deterministic assignment to provide auditable, order-invariant GST reconciliation inside client memory. Future iterations will focus on cross-tier conflict resolution and evaluation on large-scale real-world accounting records.

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